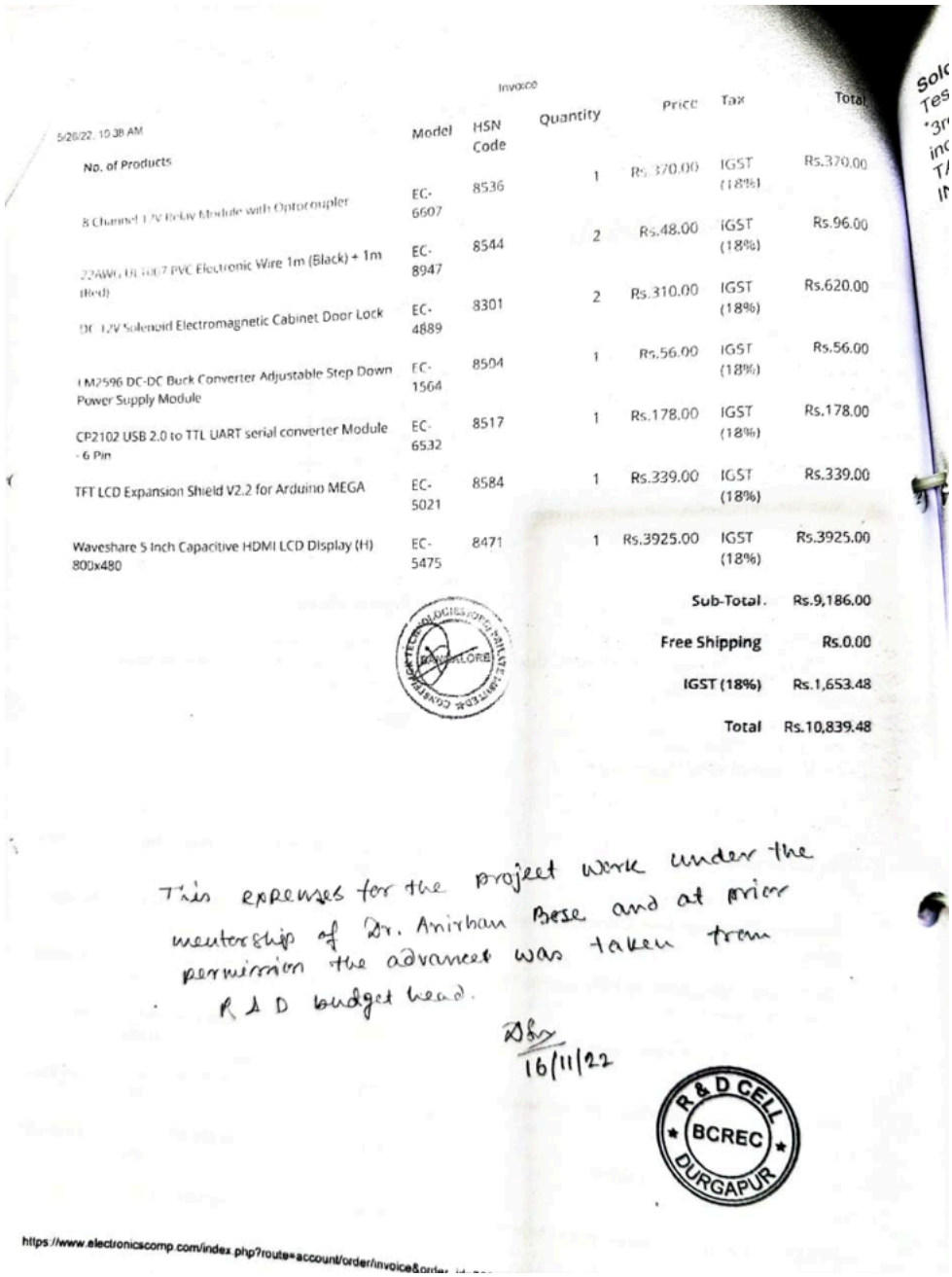


Invoice

5/26/22, 10:38 AM

No. of Products	Model	HSN Code	Quantity	Price	Tax	Total
8 Channel 12V Relay Module with Optocoupler	EC-6607	8536	1	Rs. 370.00	IGST (18%)	Rs.370.00
22AWG UL 1007 PVC Electronic Wire 1m (Black) + 1m (Red)	EC-8947	8544	2	Rs.48.00	IGST (18%)	Rs.96.00
DC 12V Solenoid Electromagnetic Cabinet Door Lock	EC-4889	8301	2	Rs.310.00	IGST (18%)	Rs.620.00
LM2596 DC-DC Buck Converter Adjustable Step Down Power Supply Module	EC-1564	8504	1	Rs.56.00	IGST (18%)	Rs.56.00
CP2102 USB 2.0 to TTL UART serial converter Module - 6 Pin	EC-6532	8517	1	Rs.178.00	IGST (18%)	Rs.178.00
TFT LCD Expansion Shield V2.2 for Arduino MEGA	EC-5021	8584	1	Rs.339.00	IGST (18%)	Rs.339.00
Waveshare 5 Inch Capacitive HDMI LCD Display (H) 800x480	EC-5475	8471	1	Rs.3925.00	IGST (18%)	Rs.3925.00

Sub-Total.	Rs.9,186.00
Free Shipping	Rs.0.00
IGST (18%)	Rs.1,653.48
Total	Rs.10,839.48

This expenses for the project work under the mentorship of Dr. Anirban Bose and at prior permission the advanced was taken from R & D budget head.

16/11/22

R & D CELL BCREC DURGAPUR

https://www.electronicscomp.com/index.php?route=account/order/invoice&order_id=...

7 FACILITIES AND TECHNICAL SUPPORT (100)

7.1 Adequate and well equipped laboratories, and technical manpower (40)

Sr. No	Name of the Laboratory	Number of students per set up(Batch Size)	Name of the Important Equipment	Weekly utilization status(all the courses for which the lab is utilized)	Technical Manpower Support		
					Name of the Technical staff	Designation	Qualification
1	Lab 1 - Ada Ba	32	Desktop Comp	CS593-OOP L/	Nayan Konar	Tech. Assistant	M.C.A.
2	Lab 2 - Turing l	32	Desktop Comp	PCC-CS494-D	Pranab Sinha	Online Progran	M.C.A.
3	Lab 3 - Aryabh:	32	Desktop Comp	CS393-PYTHC	Bappaditya Ch	Sr. Tech. Assist	P.G.D.C.A.
4	Lab 4 - Mahaln	30	Desktop Comp	CS393-PYTHC	Saikat Dey	Sr. Tech. Assist	M.C.A.
5	Lab 5 - Pascal	32	Desktop Comp	CS391-DS, ES	Pravash Das	Computer Lab .	Diploma in Cor
6	Lab 6 - Holleritl	32	Desktop Comp	ES-CS291-PR(	Kallol Chatterje	Sr. Tech. Assist	Diploma in Ele(
7	Lab 7 - Von Ne	30	Component De	PC CS392- Co	Indranil Bhatta(	Sr. Tech. Assist	DOEACC A-LE

7.2 Additional Facilities Created for Improving the Quality of Learning Experience in Laboratories (20)

Sr. No	Name of the Facility	Details	Purpose for creating facility	Utilization	Relevance to POs/PSOs
1	Machine Learning	TensorFlow tool (Python, Anaconda, MATLAB License)	Student study feature based classification and regression patterns of the data	Students create socially useful final year projects.	PO6, PO7, PO8, PO9, P10, P11, P12, PSO1, PSO3, PSO4
2	Deep Learning	Deep neural networks using NumPy (Python, Anaconda, MATLAB License)	Application of Artificial Neural Network (ANN)	Used for complex learning problems which normally happens in case of human learning	PO1, PO2, PO3, PO4, PO5, P12, PSO3, PSO4
3	Cloud Computing	Cloud based API (Python, Anaconda, OpenAIapi and tools, MATLAB License)	Provides hands on exposure of cloud computing to the students	Utilized for better career opportunities and the implementation of project	PO1, PO2, PO3, PO4, PO5, P12, PSO3, PSO4
4	Think IOT Laboratory	Dedicated Hardware Laboratory with various hardware devices like Raspberry Pi, Arduino and several IOT sensors and components	Students are provided with hands-on orientation to IOT platforms and working with various sensors to orient them to research based projects.	Students participate in different research activities and get first-hand exposure to implementing such kind of projects.	PO1, PO2, PO3, PO4, PO5, P12, PSO3, PSO4
5	Hardware Laboratory	8085/8086 kit, Logic analyzer	Students can create low level programs and various arithmetic and logical operations.	Students get knowledge of low level programming and the practical implementation of arithmetic and logical operations.	PO1, PO2, PO3, PO4, PO5, P12, PSO3, PSO4

7.3 Maintenance of laboratories and overall ambiance (10)

Maintenance of Laboratory Equipments	
Sl. No.	**Activity Description**
1.	Laboratories are well furnished.
2.	Laboratories are regularly cleaned, and systems are monitored maintained by the respective lab technician.
3.	Regular maintenance of computers and other equipment is car out periodically every semester.
4.	Dos and Don’ts and safety measures rules are displayed in all laboratories.
5.	Minor repairs are carried out by the lab technician if required.
6.	Maintenance registers are available with the third-party mainte contractor personnel deployed in the campus at the institution
7.	Maintenance and repairs are outsourced through the procedur the institution level.
8.	No one is allowed to alter or delete configuration settings of th computer system.

Overall Ambience	
Sl. No.	**Activity Description**
1.	Laboratories are regularly cleaned.
2.	The laboratories are equipped with computers and equipment t are sufficient to conduct the experiments as per the affiliating University curriculum.
3.	Sufficient numbers of windows are available for ventilation and natural light.
4.	Lighting system is very effective along with natural light in eve corner of the rooms.
5.	Power backup is available in the laboratories in case of power failure.
6.	Each Laboratory is equipped with white board, computer and internet facilities.
7.	Condition of the infrastructure is good and maintained properly
8.	Well cleaned and well-maintained work stations/equipments ar available.
9.	Laboratories are fully Air conditioned.

7.4 Safety measures in laboratories (10)

Sr. No	Laboratory Name	Safety Measures
1	Lab 1 - Ada Basic Programming Laboratory	1. Anti-skid tiles. 2. MCBs to prevent short-circuit. 3. Emergency exit. 4. Fire extinguishe
2	Lab 2 - Turing Fundamental Computing Laboratory	1. Anti-skid tiles. 2. MCBs to prevent short-circuit. 3. Emergency exit. 4. Fire extinguishe
3	Lab 3 - Aryabhatta Internetworking Laboratory	1. Anti-skid tiles. 2. MCBs to prevent short-circuit. 3. Emergency exit. 4. Fire extinguishe
4	Lab 4 - Mahalnobis Software Design Laboratory	1. Anti-skid tiles. 2. MCBs to prevent short-circuit. 3. Emergency exit. 4. Fire extinguishe
5	Lab 5 - Pascal Advanced Programming Laboratory	1. Anti-skid tiles. 2. MCBs to prevent short-circuit. 3. Emergency exit. 4. Fire extinguishe
6	Lab 6 - Hollerith Information Technology Laboratory	1. Anti-skid tiles. 2. MCBs to prevent short-circuit. 3. Emergency exit. 4. Fire extinguishe
7	Lab 7 – Von Neumnn Computer Organization Laboratory	1. Anti-skid tiles. 2. MCBs to prevent short-circuit. 3. Emergency exit. 4. Fire extinguishe
8	Lab 8 – Project Laboratory	1. Anti-skid tiles. 2. MCBs to prevent short-circuit. 3. Emergency exit. 4. Fire extinguishe
9	Lab 9 – Research Laboratory	1. Anti-skid tiles. 2. MCBs to prevent short-circuit. 3. Emergency exit. 4. Fire extinguishe

7.5 Project laboratory/research laboratory /centre of excellence (20)

Sl. No.	Facility Name
1.	Project Laboratory – Room No. 242 [62.8 sq.m.] 30 Desktop PCs connected to the internet (Core i5, 8GB RAM, Windows 11 license, python, Anaconda, MATL License)
2	Research laboratory (Room No. 239) [26.03 sq. m.] 20 WorkstationDesktop PCs connected to the internet (Core i7, 16G RAM, Windows 11 Pro license, python, Anaconda, MATLAB volume License)

8 CONTINUOUS IMPROVEMENT (80)

8.1 Actions taken based on the results of evaluation of each of the COs, POs & PSOs (40)